Newton–Raphson Load-Flow Solver

E4 234 – Advanced Power System Analysis

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1 Function and Variable Dependency Diagram

Figure 1 outlines the main computational and data flow.

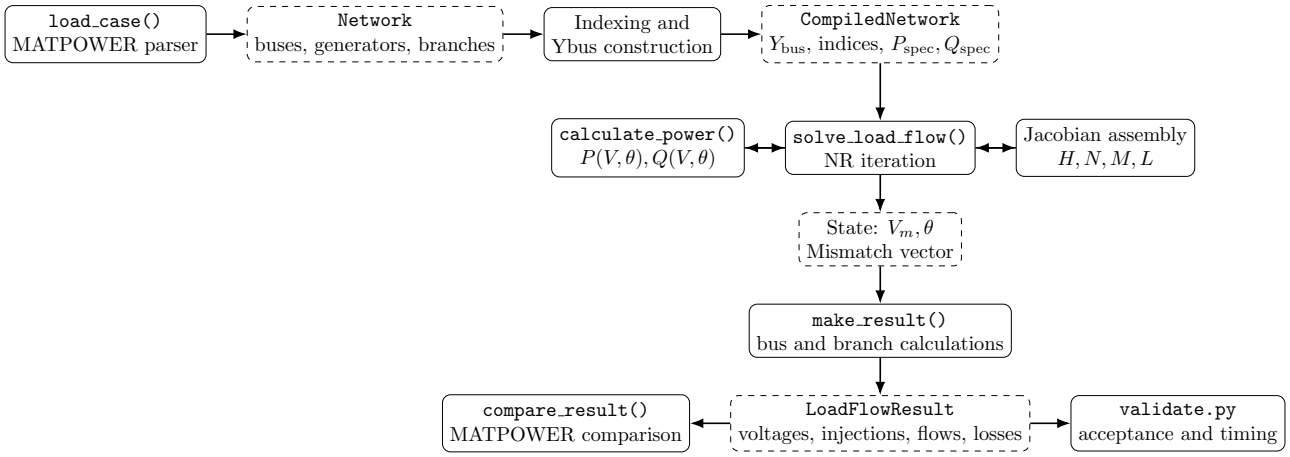


Figure 1: High-level function and variable dependency flow.

The principal numerical quantities are the bus-admittance matrix Y_{bus}, specified injections P_{spec} and Q_{spec}, voltage magnitudes V_m, voltage angles θ, calculated injections P, Q, and the active-power/reactive-power mismatch vector. The Jacobian relates the mismatch to the voltage-state correction at each Newton–Raphson iteration.

2 Simulation Configuration

Parameter	Setting
Algorithm	Newton–Raphson
Coordinate system	Polar
Initialization	Flat start
Convergence tolerance	1.0×10^{-8} pu
Maximum iterations	20
Q-limit enforcement	Disabled
Benchmark repetitions	5
Reference solver	MATPOWER

Table 1: Simulation configuration.

3 Numerical Validation and Benchmarking

The solver was compared against MATPOWER for the IEEE 14-, 30-, 57- and 118-bus cases. Acceptance limits were 10^{-4} pu for voltage magnitude error, 0.01° for angle error, 0.01 MW/MVAr for

branch-flow error and 0.1% for active-loss error. Maximum mismatch was required to be no greater than 10^{-8} pu. The IEEE 118-bus case was additionally required to converge within seven iterations.

Case	Iter.	Max mismatch (pu)	Avg. time (ms)	Time/iter. (ms)	Status
IEEE 14	4	1.030×10^{-14}	2.0670	0.5168	PASS
IEEE 30	3	9.570×10^{-10}	7.1497	2.3832	PASS
IEEE 57	4	1.542×10^{-12}	36.8960	9.2240	PASS
IEEE 118	4	1.147×10^{-10}	108.8551	27.2138	PASS

Table 2: Convergence and performance summary (five benchmark runs).

Case	V error (pu)	Angle error (°)	Branch-flow error	Active-loss error (%)
IEEE 14	6.661×10^{-16}	0.000000	0.000000	0.000000
IEEE 30	5.551×10^{-16}	0.000000	0.000000	0.000000
IEEE 57	1.554×10^{-15}	0.000000	0.000000	0.000000
IEEE 118	4.441×10^{-16}	0.000000	0.000000	0.000000

Table 3: Maximum deviations from MATPOWER. Branch-flow error is in MW/MVAr.

Case	Local solver loss (MW)	MATPOWER loss (MW)
IEEE 14	13.393272	13.393272
IEEE 30	2.443803	2.443803
IEEE 57	27.863752	27.863752
IEEE 118	132.862872	132.862872

Table 4: Active-power loss comparison.

Overall acceptance: PASS. All cases passed the stated criteria. IEEE 118 converged in four iterations, below the seven-iteration limit. The local solver and MATPOWER returned the same iteration counts and active-power losses for all four cases.

4 Detailed Newton–Raphson Results: IEEE 14-Bus Case

The case converged in four iterations with final mismatch 1.030426×10^{-14} pu. The individual solve time was 2.1668 ms; the five-run benchmark average was 2.0670 ms. The system base was 100 MVA.

4.1 Bus Results

Bus	Type	V (pu)	Angle (deg)	P_{inj} (MW)	Q_{inj} (MVAr)
1	REF	1.060000	0.000000	232.393272	-16.549301
2	PV	1.045000	-4.982589	18.300000	30.857100
3	PV	1.010000	-12.725100	-94.200000	6.075348
4	PQ	1.017671	-10.312901	-47.800000	3.900000
5	PQ	1.019514	-8.773854	-7.600000	-1.600000
6	PV	1.070000	-14.220946	-11.200000	5.230944
7	PQ	1.061520	-13.359627	0.000000	0.000000
8	PV	1.090000	-13.359627	0.000000	17.623451
9	PQ	1.055932	-14.938521	-29.500000	-16.600000
10	PQ	1.050985	-15.097288	-9.000000	-5.800000
11	PQ	1.056907	-14.790622	-3.500000	-1.800000
12	PQ	1.055189	-15.075585	-6.100000	-1.600000
13	PQ	1.050382	-15.156276	-13.500000	-5.800000
14	PQ	1.035530	-16.033645	-14.900000	-5.000000

Table 5: Solved bus voltages, angles and net injections.

4.2 Generator Input Data

The Q_g values below are initial case-file values, not solved generator reactive outputs.

Bus	P_g (MW)	Initial Q_g (MVAr)	$P_{\max}$	$P_{\min}$	Status
1	232.400	-16.900	332.400	0.000	On
2	40.000	42.400	140.000	0.000	On
3	0.000	23.400	100.000	0.000	On
6	0.000	12.200	100.000	0.000	On
8	0.000	17.400	100.000	0.000	On

Table 6: Generator input/scheduled data.

4.3 Branch Power Flows

Table 7: IEEE 14-bus branch power flows and losses (MW/MVAr).

From	To	P_f	Q_f	P_t	Q_t	Loading (%)	P_{loss}	Q_{loss}
1	2	156.882891	-20.404292	-152.585290	27.676250	0.0000	4.297600	7.271958
1	5	75.510382	3.854991	-72.747509	2.229359	0.0000	2.762873	6.084350
2	3	73.237579	3.560203	-70.914310	1.602233	0.0000	2.323269	5.162436
2	4	56.131496	-1.550350	-54.454838	3.020687	0.0000	1.676658	1.470337
2	5	41.516215	1.170998	-40.612462	-2.099034	0.0000	0.903753	-0.928036
3	4	-23.285690	4.473116	23.659135	-4.835653	0.0000	0.373445	-0.362537
4	5	-61.158230	15.823642	61.672650	-14.201005	0.0000	0.514420	1.622637
4	7	28.074176	-9.681066	-28.074176	11.384280	0.0000	0.000000	1.703214
4	9	16.079758	-0.427611	-16.079758	1.732322	0.0000	0.000000	1.304711
5	6	44.087321	12.470680	-44.087321	-8.049518	0.0000	0.000000	4.421162
6	11	7.353277	3.560473	-7.297904	-3.444514	0.0000	0.055373	0.115959
6	12	7.786067	2.503414	-7.714258	-2.353959	0.0000	0.071809	0.149455
6	13	17.747977	7.216575	-17.535891	-6.798913	0.0000	0.212085	0.417662
7	8	-0.000000	-17.162971	0.000000	17.623451	0.0000	-0.000000	0.460481
7	9	28.074176	5.778691	-28.074176	-4.976622	0.0000	-0.000000	0.802069
9	10	5.227552	4.219138	-5.214678	-4.184937	0.0000	0.012875	0.034201
9	14	9.426381	3.610006	-9.310227	-3.362931	0.0000	0.116154	0.247075
10	11	-3.785322	-1.615063	3.797904	1.644514	0.0000	0.012581	0.029451
12	13	1.614258	0.753959	-1.607960	-0.748261	0.0000	0.006298	0.005698
13	14	5.643851	1.747174	-5.589773	-1.637069	0.0000	0.054078	0.110105

The reported branch loading is zero for all branches because the result-generation code returns zero loading when a branch has no positive rating. It should not be interpreted as zero physical utilisation.

4.4 System Losses and Mismatch History

Quantity	Value
Total active-power loss	13.393272 MW
Total reactive-power loss	30.122388 MVAr

Table 8: Total system losses.

Iteration	Maximum mismatch (pu)
0	$9.21935426 \times 10^{-1}$
1	$1.00500895 \times 10^{-1}$
2	$7.10378590 \times 10^{-4}$
3	$5.97792275 \times 10^{-8}$
4	$1.03042574 \times 10^{-14}$

Table 9: Newton–Raphson mismatch history.

5 Other Test Cases

Detailed Newton–Raphson results for IEEE 30-, 57- and 118-bus cases are not repeated here. They are available in `./NR_Solver/nrlf_results.txt`.

6 Comments on AI-Assisted Solver Development

Comment 1. Initially I encountered a bug where for 118 bus system angles were not matching with reference calculations from matpower because for 118 bus system, the reference angle of voltage used was 30° (Slack Bus angle) but while running matpower solver in MATLAB, the AI generated code set all of the initial values of angle = 0, assuming a flat start. This happened because AI lost the context it was given earlier.

Comment 2. Purely AI based development of software doesn't allow the human developer to gain enough understanding of the system and debugging/adding new features becomes difficult when context is lost.